

Effects of nanosilica filler surface modification and compatibilization on the mechanical, thermal and microstructure of PP/EPR blends

Nora Gharzouli, Rachida Doufnoune, Farid Riahi & Samir Bouchareb

To cite this article: Nora Gharzouli, Rachida Doufnoune, Farid Riahi & Samir Bouchareb (2019): Effects of nanosilica filler surface modification and compatibilization on the mechanical, thermal and microstructure of PP/EPR blends, Journal of Adhesion Science and Technology, DOI: [10.1080/01694243.2018.1539157](https://doi.org/10.1080/01694243.2018.1539157)

To link to this article: <https://doi.org/10.1080/01694243.2018.1539157>



Published online: 28 Jan 2019.



Submit your article to this journal [↗](#)



Article views: 2



View Crossmark data [↗](#)



Effects of nanosilica filler surface modification and compatibilization on the mechanical, thermal and microstructure of PP/EPR blends

Nora Gharzouli^{a,b}, Rachida Doufnoune^{a,b}, Farid Riahi^c and Samir Bouchareb^{a,b}

^aUnité de Recherche des Matériaux Emergents, Sétif-1 URMES, Equipe de Valorisation des Polymères, Université Ferhat ABBAS Sétif-1, Sétif, Algérie; ^bDépartement de Génie des Procédés, Faculté de Technologie, Université Ferhat ABBAS Sétif-1, Sétif, Algérie; ^cLaboratoire des Matériaux Polymériques Multiphasiques (LMPMP), Faculté de Technologie, Université Ferhat ABBAS Sétif-1, Sétif, Algérie

ABSTRACT

Polypropylene/ethylene-propylene rubber/nanosilica (PP/EPR/nano-SiO₂) composites were prepared by a melt blending masterbatch process using a Brabender mixer. In order to improve the interfacial adhesion and achieve diverse desired properties of the composites, nanosilica surface silylation by means of two silane coupling agents: N-(β -aminoethyl)- γ -aminopropyltrimethoxysilane (AEAPTMS) and 3-methacryloxypropyltrimethoxysilane (MPTMS) was explored. The composites were also compatibilized using three compatibilizers: methyl methacrylate grafted PP (MMA-g-PP), glycidylmethacrylate grafted PP (GMA-g-PP) and maleic anhydride grafted PP (MAH-g-PP). The properties of the blends and the composites were examined using tensile and Izod impact tests, differential scanning calorimetry (DSC), thermogravimetric analysis (ATG) and scanning electron microscopy (SEM). According to the mechanical property evaluations, the incorporation of nano-SiO₂ particles into PP/EPR blend improved the tensile strength and Young's modulus of the composites. The elongation and Izod impact strength were adversely affected. A significant improvement in the mechanical properties was obtained for the composites with AEAPTMS-SiO₂ and MAH-g-PP. The DSC results indicated that the incorporation of the modified silica and MAH-g-PP increased the crystallinity of the composites. However, no significant variation in the crystallinity was observed as a result of the addition of MMA-g-PP and GMA-g-PP. The TGA results revealed that the composites exhibit a higher thermal stability than that of the neat matrix. SEM micrographs of the fractured surfaces revealed a two-phase morphology with EPR nodules being dispersed in the PP matrix. SEM also indicated that the incorporation of MAH-g-PP into PP/EPR composites contributes to a better dispersion of the EPR phase and nano-SiO₂ particles in the polymer matrix.

ARTICLE HISTORY

Received 8 July 2018
Revised 8 October 2018
Accepted 10 October 2018

KEYWORDS

PP/EPR/nanosilica ternary phase; silane coupling agents; compatibilization; composites

1. Introduction

Owing to its chemical resistance, low density, remarkable tensile property, and process versatility, polypropylene has a broad range of applications. However, the usefulness of polypropylene is limited at low temperatures by the embrittlement at its glass transition temperature (about 0 to 15 °C) [1–3]. Many attempts have been made to overcome this problem either by co-polymerisation with ethylene or melt-blending with rubbery materials. The different elastomers used include natural rubber (NR), ethylene-propylene-diene monomer (EPDM), styrene-butadiene-styrene (SBS) copolymer, ethylene-octene copolymer (EOC), nitrile butadiene-rubber (NBR), ethylene-vinyl acetate (EVA), styrene-ethylene-propylene rubber (SEP), polyisoprene (IR), polyisobutylene (PIB) and ethylene-propylene rubber (EPR) [4–10]. The automotive industry is a major growth market for this class of polymeric materials. The properties of these blends generally depend on the composition and level of dispersion of the two phases. It was established that the highest improvement in the impact strength of polypropylene was achieved by using EPR and EPDM but moderate effects were obtained from SBS and butyl rubber while IR and PIB did not affect the impact strength of polypropylene [11].

Blends of PP with EPR have been studied extensively [3,12–14]. These blends are phase separated systems in which their two-phase structures are obtained due to the lack of miscibility of the rubber and the thermoplastic phase. A pre-requisite for these materials is that the minor proportion of rubber, typically between 5 and 20% is incorporated as a disperse phase into a rigid PP matrix [3,15]. The blends show excellent processability and are characterized by high stiffness, high softening temperature, excellent low-temperature modulus and impact strength, dimensional stability, low shrinkage and good mechanical properties in a large temperature range. However, the improved low temperature impact strength of such blends may be accompanied with a reduction of other mechanical properties such as stiffness and hardness. Products with balance properties, such as good impact strength and stiffness at room and low temperature, can be obtained by the concept of a ternary blend which contains both rubber and nanofillers in the polypropylene matrix [16,17].

In such ternary systems, frequently conflicting data can be found in the literature concerning their structure and properties. Separate dispersion of the elastomer and filler in the matrix was observed, however in other studies the filler particles were reported to be encapsulated by the elastomer [18–20]. Numerous studies have revealed that both structures can be developed depending on many factors such as component properties, processing conditions, geometry and particle size of the filler [21,22]. The surface of the filler has major effects on the structure and properties of the composites. Generally, a non-treated filler is encapsulated into the elastomer phase, while the treated filler results in a good dispersion of the two components.

Among the many minerals used as reinforcing fillers in PP/elastomer blends it is worth to mention talc, kaolin, montmorillonite, different varieties of calcium carbonate, barium sulfate and natural or artificial silicas. The last two decades has seen the development of a growing interest in nanoscale fillers; among which the most common are graphene oxide (GO), carbon nanotubes (CNT), nanoclays and nanosilica (nano-SiO₂) [23–30]. The main advantage of this class of minerals is to provide, for

very low mass fractions (less than 5%), properties comparable or even superior to those of reinforced materials with 20 to 30% of conventional fillers. This paradoxical result is due to the large specific area of the nanofillers induced by their nanoscale dimensions, and the key role played by filler/filler and filler/matrix interactions. The problem, however, is the compatibility of highly hydrophilic silica with highly hydrophobic (PP/elastomer) polymers. To overcome this difficulty, both matrix and nanofiller are usually grafted with polar monomers and bifunctional silane coupling agents, respectively. The polar monomer groups are then able to interact with the functional groups on the mineral filler, while the carbon chains are able to anchor to the polymer matrix through physical entanglements and Van der Waals interactions, and thus, give a significant improvement in the properties of the silica-filled compounds [31–34].

Several studies have been reported on the PP/elastomer/silica ternary composites, while very few studies dealt with the incorporation of a compatibilizer into this type of composites systems. The microstructure and the mechanical properties of polypropylene/elastomer/silica composites were investigated by Uotila et al. [35] with the aim of improving stiffness and impact resistance. They have observed that the selectivity of silica particles for PP and EPR phases can be controlled through the selection of suitable compatibilizers. Two different microstructures were achieved in this work: separate dispersion of the filler in the polymer matrix and encapsulation of the filler by the elastomer. Chen et al. [36] investigated the effect of silica and polyurethane (PU) on the mechanical properties of PP/SiO₂-g-PU/PU hybrid composites compatibilized with H₂N-g-PP. The impact strength and ductility were significantly improved in the presence of both elastomeric PU and compatibilizer components. In accordance with Pustak et al. [37,38], the stiffness-toughness equilibrium of the PP/SiO₂/poly(styrene-*b*-ethylene-co-butylene-*b*-styrene) block copolymer (SEBS) composites were established by combining effects of stiff filler and toughened SEBS elastomer. El-Midany and Ibrahim [39] investigated the effect of SEBS and maleic anhydride grafted SEBS (MAH-g-SEBS) as compatibilizers on the mechanical properties of PP/SiO₂ binary composites. The results showed an improvement in mechanical properties after the addition of silica to the PP matrix. However, silica addition led to a drop in strain measures. On the other hand, the addition of the compatibilizer enhanced the interfacial bonding and smoothened the transfer of the stresses between filler particles and the polymeric matrix.

In this study the properties and structure of the ternary phase PP containing EPR and nano-SiO₂ are conducted. The investigation is carried out on the PP/EPR blend in the proportion of 90/10. The nano-SiO₂ particles were functionalized with two commercial silane coupling agents: N-(β -aminoethyl)- γ -aminopropyltrimethoxysilane (AEAPTMS) and 3-methacryloxypropyltrimethoxysilane (MPTMS) and then, incorporated into the matrix at concentration of 1 wt%. In order to promote the interaction between the different phases three types of compatibilizers were used, namely: methyl methacrylate grafted PP (MMA-g-PP), glycidylmethacrylate grafted PP (GMA-g-PP) and maleic anhydride grafted PP (MAH-g-PP). The compatibilizer's content into the different compositions was fixed at 20 wt%. All composites were prepared by a melt mixing process (masterbatch dilution technique) using a Brabender mixer. The

Table 1. Description of the materials used in this study.

Materials	Commercial designation	Supplier	Properties
Nano-silica	CAS No S5130	Sigma-Aldrich	$d = 2.3 \text{ g/cm}^3$, $d_{50} = 7 \text{ nm}$, pH (at 4%) = 3.7–4.7, SSA = 395 ± 25 m^2/g , purity = 99.8%
PP EPR	ISPLEN [®] PP040 G1E Dutral [®] CO 054	Repsol Quimica S.A. EniChem UK Ltd	MFI = 3 g/10 min, $d = 0.902 \text{ g/cm}^3$ $M_L (1 + 4) = 44$ at 125 °C, $d = 0.89 \text{ g/cm}^3$
MMA	Methyl methacrylate	Sigma-Aldrich	bp = 101 °C, $d = 0.94 \text{ g/cm}^3$, $M_W = 100.12 \text{ g/mol}$, purity = 97%
GMA	Glycidyl methacrylate	Sigma-Aldrich	bp = 189 °C, $d = 1.07 \text{ g/cm}^3$, $M_W = 142.15 \text{ g/mol}$, purity = 97%
MAH	Maleic anhydride	Sigma-Aldrich	bp = 202 °C, $d = 1.48 \text{ g/cm}^3$, $M_W = 98.06 \text{ g/mol}$, purity = 99%
DCP	Dicumyl peroxide	Sigma-Aldrich	bp = 31 °C, $d = 1.56 \text{ g/cm}^3$, $M_W = 270.37 \text{ g/mol}$, purity = 98%
AEAPTMS	Dow Corning [®] Z-6020	Sigma-Aldrich	bp = 263 °C, $d = 1.02 \text{ g/cm}^3$, M_W = 222.36 g/mol, purity = 98%
MPTMS	Silquest A-174	Sigma-Aldrich	bp = 190 °C, $d = 1.045 \text{ g/cm}^3$, $M_W = 248.35 \text{ g/mol}$, purity = 98%

mechanical properties (tensile strength and impact resistance) of the prepared materials were studied. The thermal properties were characterized by using differential scanning calorimetry (DSC) and thermogravimetric analysis (TGA). The microstructures of the samples were analyzed by scanning electron microscopy (SEM).

2. Experimental

2.1. Materials

A full description of the materials used is given in Table 1. The polymer-matrix used in this study was a commercial isotactic polypropylene homopolymer (iPP) (ISPLEN PP040 G1E) supplied by Repsol Quimica S.A. (Spain). The melt flow index (MFI) and density were 3 g/10 min (230 °C/2.16 kg, ISO1133) and 0.905 g/cm³ (ISO 1180), respectively. The rubbery modifier for iPP, Ethylene-Propylene Rubber (Dutral[®] CO 054) was a commercial grade and purchased from EniChem (USA). Its polypropylene content, Mooney Viscosity (ML 1 + 4, 125 °C) and density were 41 wt%, 44 and 0.89 g/cm³, respectively. Amorphous pyrogenic silica, referenced as CAS No S5130 was obtained from Sigma-Aldrich. The silica treatment reagents used: N-(β -aminoethyl)- γ -aminopropyltrimethoxysilane (AEAPTMS) and γ -methacryloxypropyltrimethoxysilane (MPTMS) were purchased from Sigma-Aldrich. The chemical structures of the silane coupling agents are given in Figure 1. The reagents: methylmethacrylate (MMA), glycidylmethacrylate (GMA), maleic anhydride (MAH) monomers and the free radical initiator, dicumyl peroxide (DCP), used for the synthesis of the compatibilizers were supplied by Sigma-Aldrich. All other reagents which were laboratory grades and supplied from Sigma-Aldrich were used without further purification.

2.2. Surface silylation of nanosilica

The silylation process of nano-SiO₂ with silanes was carried out in a solution of methanol/water at a ratio of 50:50 (v/v) at pH = 3 by using acetic acid. About 10 wt%

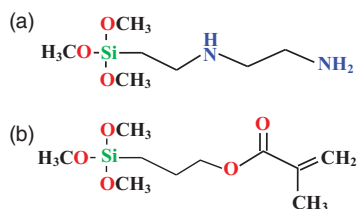


Figure 1. Chemical structures of the silane coupling agents (a) AEAPTMS and (b) MPTMS.

of each silane was introduced in 300 ml of the mixture and stirred vigorously with a magnetic stirrer for 3 h at 60 °C. Then, 2 g of nano-SiO₂ were added to the solution and the grafting reaction was achieved under vigorous stirring during 5 h at 80 °C. The product was then separated by filtration and washed with methanol then with hot water repeatedly in order to remove the excess silane. The modified product was dried in a vacuum oven at 120 °C for 3 h, then ground in an agate mortar to pass a 200 mesh sieve and stored for further use. The silylated samples prepared from the silanes were denoted hereinafter in the manuscript as AEAPTMS-SiO₂ and MPTMS-SiO₂.

2.3. Preparation of compatibilizers

The grafting reactions onto PP were carried out in the molten state using a 50 mL compounding chamber of a PlastographTM PL2100 internal mixer (Brabender®, Duisbourg, Germany) and DCP as a free radical initiator. In a typical procedure, PP, 2 wt% of monomers, and 0.1 wt% of DCP were introduced into the preheated mixing chamber, and the grafting reaction was performed at 200 °C for 8 min at 200 °C at constant mixing speed of 30 rpm. At the end of the reaction, the product was removed from the plastograph chamber, cooled, solidified and granulated into pellets and purified in boiling xylene for 24 h to remove any unreacted monomer. Finally, the grafted PP was vacuum dried at 60 °C to constant weight. The grafted PP prepared from MMA, GMA and MAH were denoted as MMA-g-PP GMA-g-PP and MAH-g-PP, respectively.

2.4. Elaboration of composites

The composites were prepared in a two-step process. First, masterbatches were produced by dispersing pristine nano-SiO₂ or functionalized nano-SiO₂ (AEAPTMS-SiO₂ and MPTMS-SiO₂) into the compatibilizer using the Brabender[®] mixer at 190 °C for 12 min and a rotating speed of 30 rpm. Next, the masterbatches were mixed with PP/EPR blends (90/10 w/w) using the Brabender[®] internal mixer for 10 min at 200 °C and a rotating speed of 40 rpm. Formulations were prepared such that all PP/EPR/SiO₂ composites contained 1 wt% and 20 wt% of nano-SiO₂ and compatibilizers, respectively. The prepared compositions were pelletized and compression molded into standard test specimens for tensile and impact test using a hydraulic press operating at 200 °C. After heating for 2 min, the samples were sustained to 190 Kg/cm² pressing and allowed to stand this pressure for 4 min. Finally, the samples were slowly

Table 2. Composition and codes of PP/EPR/nanosilica composites.

Formulations	PP/EPR (%)	MMA-g-PP (%)	GMA-g-PP (%)	MAH-g-PP (%)	SiO ₂ (%)	AEAPTMS-SiO ₂ (%)	MPTMS-SiO ₂ (%)	Codes
PP	100	0	0	0	0	0	0	F0
PP/EPR	100	0	0	0	0	0	0	F1
PP/EPR/MMA-g-PP	80	20	0	0	0	0	0	F2
PP/EPR/GMA-g-PP	80	0	20	0	0	0	0	F3
PP/EPR/MAH-g-PP	80	0	0	20	0	0	0	F4
PP/EPR/SiO ₂	99	0	0	0	1	0	0	F5
	99	0	0	0	0	1	0	F6
	99	0	0	0	0	0	1	F7
PP/EPR/MMA-g-PP/SiO ₂	79	20	0	0	1	0	0	F8
	79	20	0	0	0	1	0	F9
	79	20	0	0	0	0	1	F10
PP/EPR/GMA-g-PP/SiO ₂	79	0	20	0	1	0	0	F11
	79	0	20	0	0	1	0	F12
	79	0	20	0	0	0	1	F13
PP/EPR/MAH-g-PP/SiO ₂	79	0	0	20	1	0	0	F14
	79	0	0	20	0	1	0	F15
	79	0	0	20	0	0	1	F16

cooled down in the open air to room temperature. Table 2 summarizes all the prepared mixtures with their respective designations.

3. Characterization

3.1. FT-IR spectroscopy

All FT-IR spectra of the samples were recorded in the range $4000\text{--}400\text{ cm}^{-1}$ at a constant spectral resolution of 4 cm^{-1} and an averaged signal from 16 scans. Neat and treated nano-SiO₂ samples were compacted to thin pellets using a hydraulic press and then FTIR spectra were recorded in the attenuated total reflectance mode using a JASCO-IR spectrometer. To confirm the grafting reactions of MMA, GMA and MAH onto PP, a Shimadzu 8400S spectrometer was employed to acquire FTIR spectra from samples compressed to thin films using a hydraulic press operating at 200°C and 180 kg/cm^2 .

3.2. Mechanical properties

Tensile properties of the studied materials were determined at room temperature in compliance with ASTM D638-14 ($165 \times 13 \times 3.2\text{ mm}$) specifications using an Instron MTS 500 instrument at a crosshead speed of 50 mm/min and a fixed gauge length of 50 mm ($1/\text{min}$ strain rate). Tensile strength, percentage elongation at break and tensile modulus values were recorded. The notched Izod impact strength measurements were acquired in compliance with ASTM D256-88 ($80 \times 12.7 \times 3.17\text{ mm}$) using a Ceast 9050 impact pendulum tester equipped with a hammer of 1.8 Kg delivering an impact energy of 7.5 kJ at a speed of 2.88 m/s . The notch was milled with a

45° ± 1° angle and 2.5 mm depth with an automatic apparatus (Ceast Notchvis). All measurements were made in five replicates and the values were averaged.

3.3. DSC analyses

Representative samples were analyzed using a Netzsch thermal Analyzer differential scanning calorimeter following a thermal cycling procedure. The instrument was first calibrated using indium as a reference material. DSC analyses of the samples were performed under a nitrogen atmosphere. The examined sample weights ranged from 10 to 15 mg and all materials were analyzed at the same scan rate. Melting and crystallization temperatures were recorded. The samples were heated to 200 °C at a rate of 10 °C/min and held at 200 °C for 5 min to eliminate the influence of thermal history. Then, the samples were cooled to 0 °C at a rate of 10 °C/min, and then heated again to 200 °C at the same rate. The temperature and heat flow scales were calibrated using high purity indium and zinc samples. The melting curves of the second run were recorded. The melting temperatures (T_m) were taken as the temperature corresponding to the peak values of the melting endotherm. The degree of crystallinity was calculated as the ratio of the melting enthalpies divided by the weight fraction of PP (w_{PP}) in the blend.

$$X_c = \frac{\Delta H_f^{obs}}{w_{PP} \Delta H_f^0} \cdot 100$$

Where ΔH_f^{obs} is the measured enthalpy of melting, ΔH_f^0 is the ideal melting enthalpy of a perfect PP crystal. A value of $\Delta H_f^0 = 209 \text{ J g}^{-1}$ for 100% crystalline PP homopolymer was used for calculating the X_c [40].

3.4. TGA analyses

The degradation behavior of the polymer composites samples was evaluated using a Netzsh STA 449C thermal analyzer apparatus. The test was carried out in nitrogen atmosphere by increasing the temperature from room temperature to 580 °C at a heating rate of 10 °C/min.

3.5. SEM analyses

The microstructure of samples was examined by scanning electron microscopy (SEM) by using a Quanta 200 Scanning Electron Microscope at an acceleration voltage of 10 kV. Prior to observations, surfaces of the samples were cryogenically fractured in liquid nitrogen to avoid any plastic deformation.

4. Results and discussion

4.1. Characterization of nano-SiO₂ silylated with AEAPTMS and MPTMS

Figure 2 shows the spectra of the neat nano-SiO₂ particles, the AEAPTMS-SiO₂ and MPTMS-SiO₂ samples. In Figure 2(a) the characteristic broad band centered at

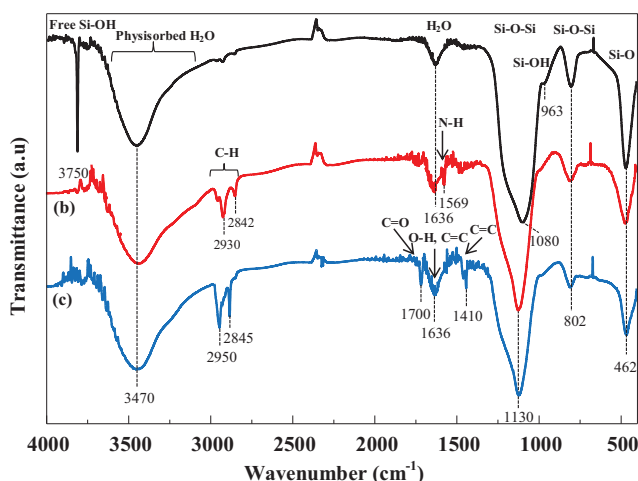


Figure 2. FT-IR spectra of (a) neat nano- SiO_2 , (b) AEAPTMS- SiO_2 and (c) MPTMS- SiO_2 .

3470 cm^{-1} is due to the hydroxyls of physisorbed water molecules and the silanol groups of the silica surface (vicinal OH). The sharp absorption band observed at 3750 cm^{-1} is ascribed to free hydroxyl groups and the weak band observed at 963 cm^{-1} is assigned to the surface free Si-OH groups (isolated and geminated OH). The absorption band at 1636 cm^{-1} is ascribed to the adsorbed water molecules with their hydroxyl groups. The strong absorption band observed at 1080 cm^{-1} is assigned to Si-O-Si asymmetric stretching vibration and the absorption bands at 802 and 462 cm^{-1} are attributed to Si-O-Si and Si-O symmetric stretching vibration and bending mode, respectively [41,42]. The spectrum of AEAPTMS- SiO_2 (Figure 2(b)) exhibits two strong bands at 2930 cm^{-1} and 2842 cm^{-1} , which are attributed to the asymmetric and symmetric stretching bands of the CH_2 groups of aminopropyl and aminoethyl, respectively. The observed band at 1569 cm^{-1} can be ascribed to the N-H bending vibration of primary amines. The band at 1130 cm^{-1} assigned to Si-O-Si stretching vibration of the polysiloxane structure with a relatively high-molecular-weight, indicates the deposition of the silane coupling agent on the nano- SiO_2 particles surface. The adsorption of the aminofunctional silanes in the form of high-molecular-weight structures was also confirmed in other investigations. Studies presented in the literature indicate that the adsorption of the aminofunctional silanes in the form of high-molecular-weight structures is presumably due to the catalytic role of the amino group in the adsorption process [43,44]. In the MPTMS- SiO_2 spectrum (Figure 2(c)), the same band at 1130 cm^{-1} assigned to the Si-O-Si structure is clear. The similarity with the FTIR spectrum of AEAPTMS- SiO_2 confirms that the same structure is formed at the surface of the nano- SiO_2 particles. The bands positioned at 2950 cm^{-1} and 2849 cm^{-1} are attributed to the propyl chains of the MPTMS. The bands at 1636 and 1410 cm^{-1} are attributed to C=C stretching of methacrylate and vinyl groups, respectively. The band which appears at 1700 cm^{-1} is ascribed to the vibration of the ester carbonyl groups C=O of methacryloxysilane [45]. These results confirm the successful grafting of nano- SiO_2 with MPTMS coupling agent.

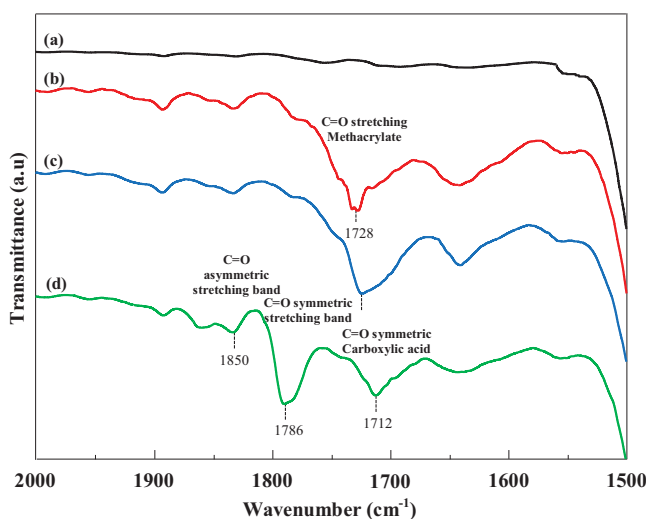


Figure 3. FT-IR spectra in the range $2000\text{--}1500\text{ cm}^{-1}$ of (a) PP, (b) MMA-g-PP, (c) GMA-g-PP and (d) MAH-g-PP.

4.2. Characterization of grafted polypropylene

Figure 3 shows the FT-IR spectra of neat PP and MMA-g-PP, GMA-g-PP and MAH-g-PP. While no absorption bands were observed in the range between $2000\text{--}1500\text{ cm}^{-1}$ for pure PP (Figure 3(a)), the modified PP grafted with MMA, GMA and MAH (Figures 3(b)–3(d)) were found to exhibit new bands, indicating the occurrence of grafting reactions. In Figure 3(b), the FT-IR spectrum of MMA-g-PP presents a strong characteristic band at 1728 cm^{-1} (C=O symmetric stretching band) which indicates the presence of carbonyl groups. The spectrum of GMA-g-PP (Figure 3(c)) shows a band at 1723 cm^{-1} , which is characteristic of the carbonyl groups of GMA. The FT-IR spectrum of MAH-g-PP (Figure 3d)) revealed three bands at 1850 , 1786 and 1712 cm^{-1} . The first two bands are attributed to cyclic anhydride groups and the third one is assigned to the carboxyl groups of an acid function. According to the literature, this result is justified by the opening of the maleic anhydride ring because of its great susceptibility to hydrolysis [33].

4.3. Mechanical properties

Mechanical properties of composite materials depend on the characteristics of both matrix and filler and on the matrix/filler adhesion in the interfacial region. In this study the effect of the addition of compatibilizers, elastomer, non-modified and silane modified nano-SiO₂ fillers on the tensile properties of the PP/EPR composites were investigated. The effect of adding EPR and compatibilizers on the tensile and impact properties of polypropylene are reported in Figure 4. The results showed that the incorporation of the EPR copolymer phase led to an important improvement in ductility of PP matrix. This was manifested by the elongation at break and impact strength of PP which increased by 300% and 276%, respectively. The increased ductility could be explained by enhanced toughening or plastification with the presence of

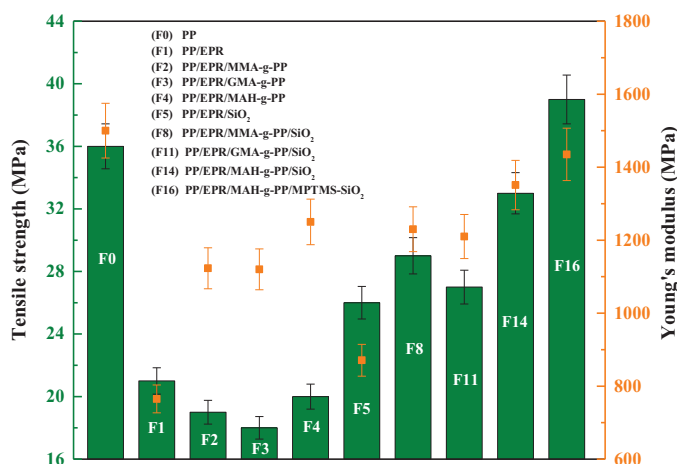


Figure 4. Tensile strength and Young's modulus of PP, PP/EPR, compatibilized blends and composites.

the elastomer. In addition, it is noted that the incorporation of EPR into PP has decreased the tensile strength and Young's modulus by 42% and 4%, respectively. These observations are in accordance with the results reported for classical polymer-elastomer blends and have been noted to be attributed to the softening or diluting action of the second phase [46,47].

The effect of compatibilizers on the mechanical properties of PP/EPR blend is also presented in Figures 4 and 5. It was found that the addition of compatibilizers in PP/EPR blend has slightly decreased the tensile strength, significantly reduced the elongation at break and impact strength (Figure 4), while considerably improved the Young's modulus (Figure 5). For PP/EPR blend containing MAH-g-PP compatibilizer, the Young's modulus increased by 63%, as compared to neat PP/EPR blend. This improvement is attributed to the increased interfacial adhesion between the PP and EPR phases and owing to the reduced particle size of the EPR dispersed phase.

As shown in Figure 4 the incorporation of the nanofiller has substantially improved tensile strength and Young's modulus of PP/EPR blend. The tensile strength and Young's modulus increased by about 24% and 14%, respectively as a result of the addition of nano-SiO₂ to PP/EPR blend. The increase in tensile strength and Young's modulus is however at the expense of elongation at break and impact strength (Figure 5). It can be seen that the elongation at break and impact strength decreased by about 21% and 6% respectively, as compared to the neat PP/EPR blend. This is due to the fact that the polymer matrix has low polarity; paradoxically, the nanosilica surface has polar hydroxyl groups and are incompatible only with a non-polar polymer.

The addition of compatibilizers to the composites in order to mediate the polarity between the nano-SiO₂ surface and the polymer matrix led to non-polar macromolecule segments of PP/EPR able to interact with the surface of nanosilica. The results showed that the tensile strength and Young's modulus of PP/EPR/MMA-g-PP/SiO₂ increased by about 12% and 42%, those of PP/EPR/GMA-g-PP/SiO₂ increased by about 4% and 39% and those of PP/EPR/MAH-g-PP/nano-SiO₂ increased by only

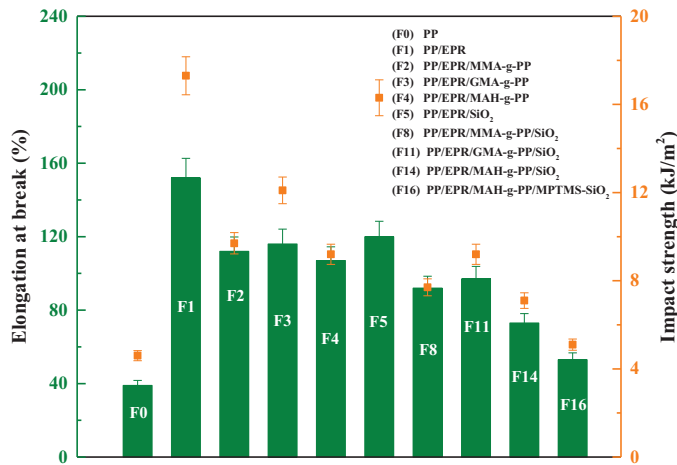


Figure 5. Elongation at break and impact strength of PP, PP/EPR, compatibilized blends and composites without silane coupling agents.

27% and 55%, respectively as compared to the uncompatibilized composites. The composites with MAH-g-PP exhibited somewhat higher tensile strength and Young's modulus values than those of the composites with MMA-g-PP and GMA-g-PP. These differences could be caused by somewhat stronger interactions between MAH-g-PP/SiO₂ than those between respective pairs of MMA-g-PP/SiO₂ and GMA-g-PP/SiO₂ systems. It is interesting to note that the MAH-g-PP compatibilized composites showed lower elongation at break and impact strength as compared to MMA-g-PP and GMA-g-PP compatibilized composites. As shown in Figure 5, the elongation at break and impact strength decreased greatly from 120% and 16.3 kJ/m² for the uncompatibilized composite to 73% and 7.1 kJ/m², which corresponds to a decrease of about 39% and 56%, respectively after the incorporation of MAH-g-PP into the composites.

Figure 6 shows the plots of the tensile strength and Young's modulus of the uncompatibilized and compatibilized composites in presence of silylated nano-SiO₂. The addition of AEAPTMS-SiO₂ and MPTMS-SiO₂ clearly increased the tensile strength and the Young's modulus of the uncompatibilized composites. At the same time the composites became much weaker, as can be seen by the lower values of elongation at break and impact strength compared to those obtained with the composites in the absence of silane coupling agents. Addition of compatibilizers to the composites led to significant improvements in tensile strength and Young's modulus. Higher values of tensile strength and Young's modulus were observed for the composites with AEAPTMS-SiO₂ and compatibilized with MAH-g-PP. For this sample, the tensile strength and Young's modulus increased by about 32% and 52%, respectively compared to the composites without compatibilizer. Such improvement is due to the presence of strong interfacial interaction between polymer chains and silica nanoparticles, leading to a better interfacial stress transfer efficiency. Doufnoune and Haddaoui [3] have suggested that the anhydride group react with the amino groups anchored on AEAPTMS surface, resulting in an improved compatibility of the nano-filler with the polymer matrix. Similar improvements in tensile strength and Young's

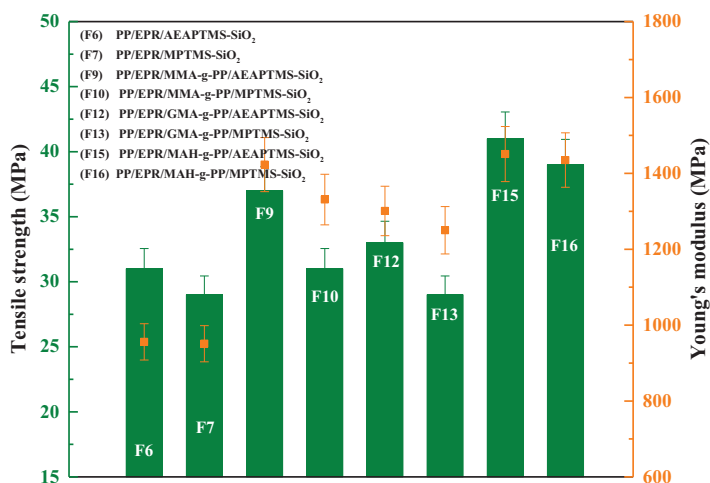


Figure 6. Tensile strength and Young's modulus of uncompatibilized and compatibilized composites without and with silane coupling agents.

modulus were also observed with all compatibilized materials, but to a lesser degree than the composite with MAH-g-PP.

The elongation at break and impact strength (Figure 7), however, decreased much with the incorporation of compatibilizers. When MAH-g-PP was used as compatibilizer, the elongation at break as well as the impact strength for the composite with AEAPTMS decreased by about 58% and 64%, respectively compared to the same system without compatibilizer. An explanation for this is that there are strong interactions between the different phases of the composites.

Figure 8 depicts the typical stress-strain curves of PP, PP/EPR blend and its composites compatibilized with MAH-g-PP and reinforced with pristine and silylated nano-SiO₂. The results revealed that the incorporation of silylated nano-SiO₂ has substantially improved the tensile strength as well as the Young's modulus. The reinforcement is more pronounced for the composites with AEAPTMS-SiO₂ and compatibilized with MAH-g-PP. The tensile strength and Young's modulus were 19 MPa and 756 MPa, respectively for PP/EPR/AEAPTMS-SiO₂ composite, whereas their values for PP/EPR/MAH-g-PP/AEAPTMS-SiO₂ composite increased by 116% and 92%, respectively, as compared to the uncompatibilized composite. At the same time, it was observed that the elongation at break and the impact strength for the same composite decreased dramatically by approximately 58% and 64%, respectively, as compared to the composite without compatibilizer. Consequently, the decrease of these properties can be attributed to the restriction of mobility of the polymer chains as a result of strong interfacial interactions [33,48].

This variation mostly depends on the interface morphology and the microstructure of the composites. Several studies have attributed the reinforcement obtained in polymer-nanofiller composites to the effectiveness of stress transfer between matrix and filler. For well-bonded nanoparticles, the applied stress can be efficiently transferred to the nanoparticles in the matrix, which clearly improves the tensile strength and Young's modulus. Nevertheless, for poorly bonded nanoparticles, the strength

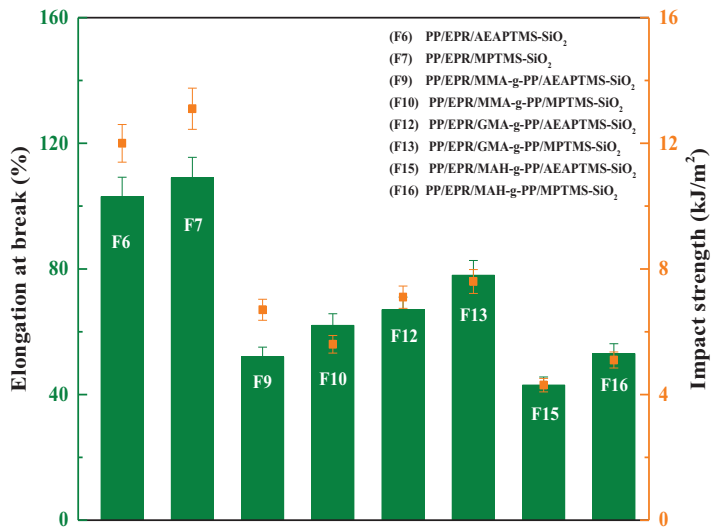


Figure 7. Elongation at break and impact strength of uncompatibilized and compatibilized composites without and with silane coupling agents.

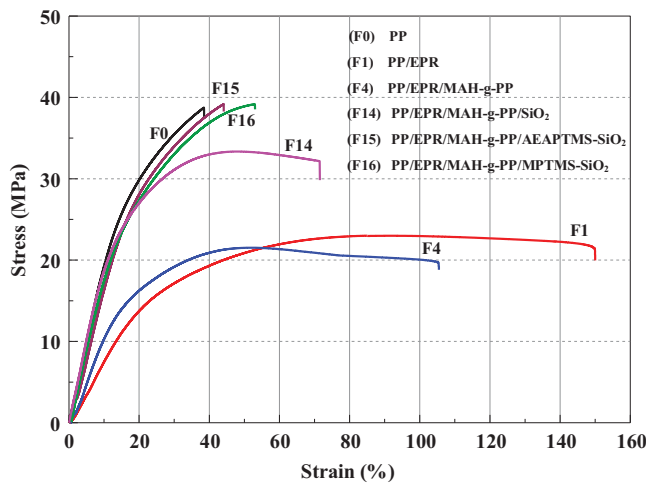


Figure 8. Stress-strain curves of PP, PP/EPR blend and its corresponding composites without and with MAH-g-PP and silane coupling agents.

decrease occurs upon adding nanoparticles because the adhesion is very weak at the interfacial region.

4.4. Differential scanning calorimetry (DSC)

Figure 9 shows the crystallization exotherms and the fusion endotherms of unmodified PP, PP/EPR blend and its composites with and without MAH-g-PP in the temperature range from 100 to 200 °C. The melting temperature (T_m), crystallization temperature (T_c), crystallinity content (X_c) and heat of fusion (ΔH) values for all samples are given in Tables 3 and 4. As seen in Figure 9 and Table 3, the thermograms showed that all

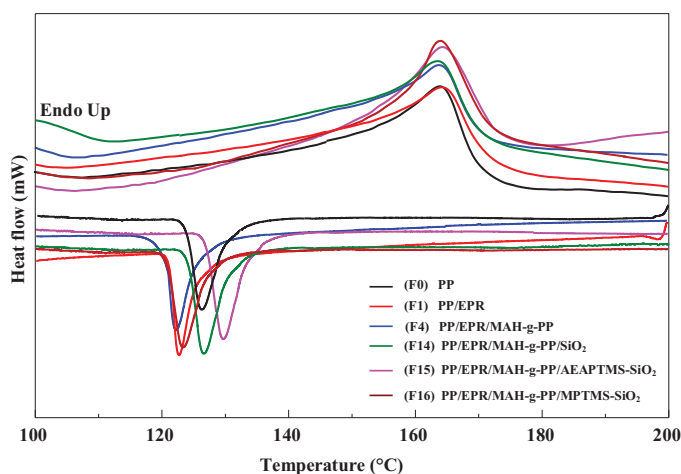


Figure 9. Heating and cooling thermograms of PP, PP/EPR blend and its corresponding composites without and with MAH-g-PP and silane coupling agents.

materials exhibited a single crystallization exotherm and a single melting endotherm. As can be seen, the T_m peak of neat PP is observed at 164 °C and remained unchanged with the addition of EPR and compatibilizers. Also, the incorporation of both pristine and silylated nanofiller did not induce any noticeable change in the T_m , and the slight deviations observed in temperature fell within margin for error. However, addition of EPR to PP resulted in a marked change in the crystallization behavior of the blend. The T_c peak of PP is 126 °C, while the EPR addition led to its decrease to 122 °C. This decrease in T_c peak not only implies a reduction in spherulite size but also a reduced thickness of crystalline lamellae. The addition of the compatibilizers to PP/EPR blend resulted in a slight decrease of T_c peak (~ 2 °C). The presence of the compatibilizer entities located at the interface resulted in a decrease of EPR dispersed phase domains, which limited the crystallization potential of PP continuous phase. The incorporation of both pristine and silylated nano-SiO₂ particles in the PP/EPR system did not cause any change in the T_c peak. Addition of the compatibilizers to the composites resulted in a noticeable increase in T_c peak. As shown in Table 7, the T_c increased from 120 °C for the uncompatibilized composite to 127 °C for the compatibilized composite with MAH-g-PP. Nevertheless, the presence of MMA-g-PP and GMA-g-PP compatibilizers, led to a slight increase in T_c peak of the composites. The increase in T_c when MAH-g-PP was used as compatibilizer is due to increased interaction between the different phases. In addition, the T_c increased from 127 °C for the composite with unmodified nano-SiO₂ to 130 °C for the composite with AEAPTMS-SiO₂. We speculate that there are strong interactions between the NH₂ groups of aminosilane on the surface of the nanofiller and the anhydride groups of the MAH-g-PP. Therefore, the combination of AEAPTMS-SiO₂ and MAH-g-PP can effectively improve the interfacial compatibility and the dispersion of the nanoparticles in the matrix. As a result, the nanofiller, has a greater surface area in the polymer matrix, which induces the nucleation of the polymer matrix. These observations support the conclusions from mechanical properties findings.

Table 3. DSC analysis data for neat PP, PP/EPR, PP/EPR/nanosilica, compatibilized blends and composites.

Materials	Melting temperature T_f ($^{\circ}\text{C}$)	Crystallization Temperature T_c ($^{\circ}\text{C}$)	Heat of Fusion ΔH (J/g)	Crystallinity Content X_c (%)
PP	164.0	126	94.9	45.4
PP/EPR	164.0	122	73.3	39
PP/EPR/MMA-g-PP	163.6	120.1	51.6	35.2
PP/EPR/GMA-g-PP	163.7	119.8	52.1	35.6
PP/EPR/MAH-g-PP	163.9	120.6	54.3	37.1
PP/EPR/SiO ₂	163.7	120.0	69.1	37.1
PP/EPR/MMA-g-PP/SiO ₂	163.2	119.2	56.7	39.3
PP/EPR/GMA-g-PP/SiO ₂	163.9	118.7	57.4	39.8
PP/EPR/MAH-g-PP/SiO ₂	164.1	127.0	52.2	41.2

On the other hand, the area of the melting endotherm was calculated and reported as the heat of fusion, ΔH . The latter is an important parameter since its magnitude is directly proportional to the overall level of X_c possessed by the polymer [49]. Blending PP with EPR correspondingly results in a decrease in X_c of PP, as evident in Table 6. The results showed that X_c of the PP decreased from 45% to 39% in the presence of EPR. Upon addition of compatibilizers to the PP/EPR system the crystallinity of PP was altered. It should be noted that the presence of compatibilizer restricts crystallizability of the polymer due to the same reasons mentioned above.

Concerning the uncompatibilized composites, silica nanoparticles are ineffective because they decreased PP crystallizability. For the case of compatibilized composite systems, the nanofiller acts as a nucleating agent for the crystallization of the PP matrix and increases the crystallinity content of the PP matrix. It is interesting to note that the presence of MAH-g-PP in the composite strongly affects the X_c of PP. The results showed that X_c of the PP increased from 37% for the uncompatibilized composite up to 44.7% for the PP/EPR/MAH-g-PP/AEAPTMS-SiO₂ system (Table 4). This indicates that not only nano-SiO₂, but also the anhydride groups of MAH-g-PP have a heterogeneous nucleation effect on PP macromolecular segments. Similar results have been reported for other three-phase PP/elastomer/filler systems [50,51]. In contrast to AEAPTMS-SiO₂ and MAH-g-PP, no significant variation in crystallinity was observed with the addition of MMA-g-PP and GMA-g-PP in the composites.

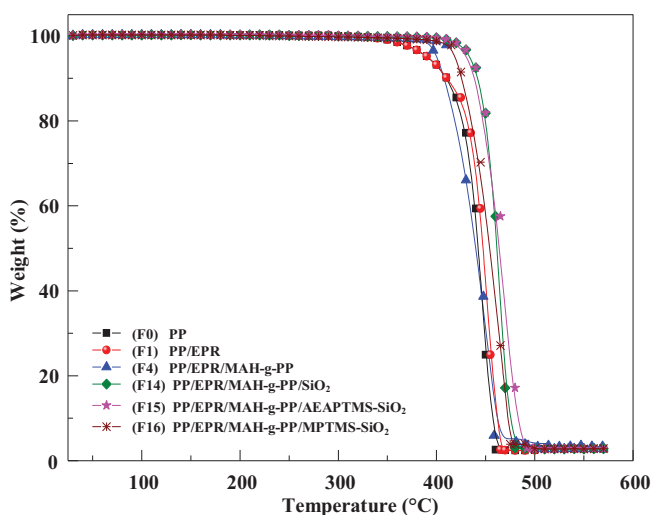
4.5. Thermogravimetric analysis (TGA)

Figure 10 shows the result of the thermogravimetric analysis of neat PP, blends and composites with and without MAH-g-PP and coupling agents. During the thermal degradation, the TGA curves displayed a single step degradation process for all the samples. Addition of elastomer and compatibilizer does not induce any effect on the decomposition process in comparison to neat PP. From the TGA curves, we have extracted two characteristic temperatures. The first of which is the temperature of the onset of decomposition (T_{onset}) that takes place at 365 $^{\circ}\text{C}$. The second is the one that marks the constant weight reached that corresponds to the maximum temperature of decomposition (T_{max}) around 462 $^{\circ}\text{C}$.

In the case of PP/EPR/MAH-g-PP/SiO₂, the T_{onset} and T_{max} shifted to 400 $^{\circ}\text{C}$ and 474 $^{\circ}\text{C}$, corresponding to an increase by 25 $^{\circ}\text{C}$ and 12 $^{\circ}\text{C}$, respectively compared to

Table 4. DSC analysis data for PP/EPR/nanosilica with silane coupling agents and compatibilized composites.

Materials	Melting temperature T_f ($^{\circ}\text{C}$)	Crystallization temperature T_c ($^{\circ}\text{C}$)	Heat of fusion ΔH (J/g)	Crystallinity content X_c (%)
PP/EPR/SiO ₂				
AEAPTMS	164.1	120.0	48.6	33.7
MPTMS	163.6	120.1	51.0	35.4
PP/EPR/MMA-g-PP/SiO ₂				
AEAPTMS	162.9	119.1	53.6	37.2
MPTMS	163.2	118.2	56.4	39.1
PP/EPR/GMA-g-PP/SiO ₂				
AEAPTMS	163.7	117.8	59.1	41.0
MPTMS	164.0	117.4	57.1	39.6
PP/EPR/MAH-g-PP/SiO ₂				
AEAPTMS	164.3	130.0	61.7	44.7
MPTMS	163.7	122.1	61.0	42.3

**Figure 10.** TGA curves of PP, PP/EPR blend and its corresponding composites without and with MAH-g-PP and silane coupling agents.

neat PP, PP/EPR and PP/EPR/MAH-g-PP blends. When the silylated SiO₂ nanoparticles are incorporated into the matrix the T_{onset} and T_{max} shifted to 430 $^{\circ}\text{C}$ and 480 $^{\circ}\text{C}$, with an increase of 30 $^{\circ}\text{C}$ and 6 $^{\circ}\text{C}$ respectively, compared to the PP/EPR/MAH-g-PP/SiO₂ system. This improvement in the thermal stability may be attributed to the fact that silylated SiO₂ nanoparticles could induce carbonization and form a thermally stable-char, and therefore reduce the releasing rate of degradation volatiles [52,53]. The reason of the improvement of the thermal stability can also be attributed to the better adhesion between the polymer matrix and the nanofiller owing to the compatibilizer and silane coupling agent's effects.

4.6. Morphology analysis (SEM)

Figure 11 shows the SEM micrographs of neat PP, PP/EPR blend without and with MAH-g-PP. Figure 11(a) shows that the fractured surface of PP matrix is smoother

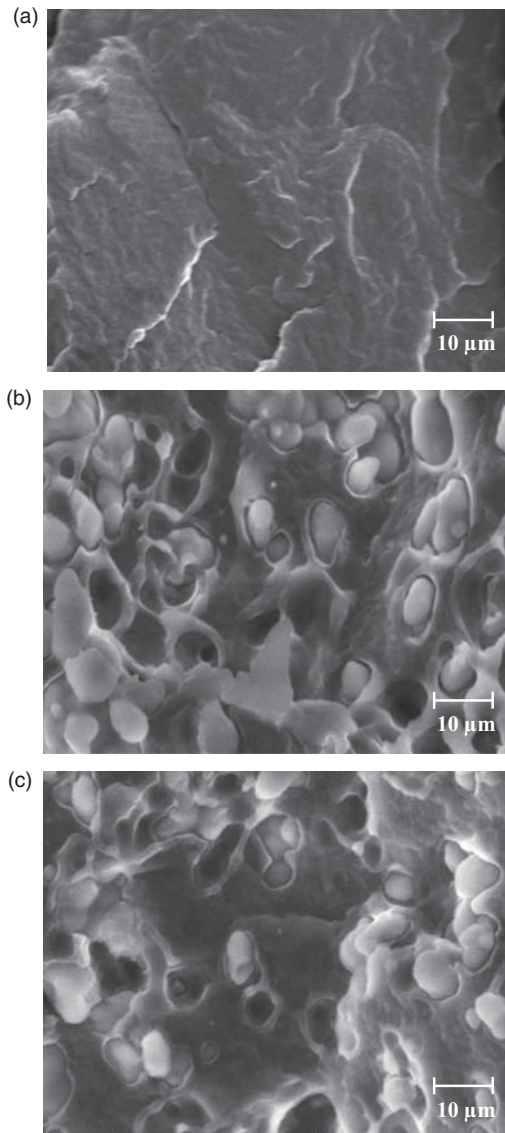


Figure 11. SEM micrographs of (a) PP, (b) PP/EPR and (c) PP/EPR/MAH-g-PP.

featuring a brittle fracture. The SEM micrograph of PP/EPR blend presented in [Figure 11\(b\)](#) shows two distinct phases and the nodules of EPR dispersed as spherical domains in the continuous PP matrix. This is due to the higher melt viscosity and lower EPR content compared to PP in the blend. In addition, the formation of EPR particles results in an interface between EPR and PP, which results in a phase separation in PP/EPR blend that is in relation to its mechanical properties. On the other hand, the empty cavities observed in the PP/EPR blend are believed to be due to the elastomeric particles dislodged from the fractured surface, reflecting a poor adhesion to the matrix. From [Figure 11\(c\)](#), it is observed that the presence of the MAH-g-PP in PP/EPR blend does not induce visible changes in the EPR nodule sizes compared

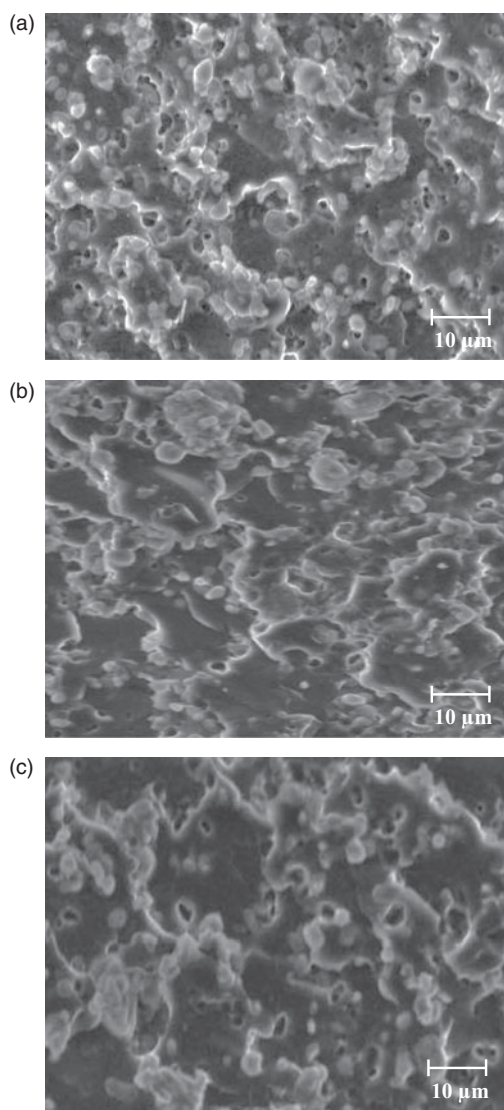
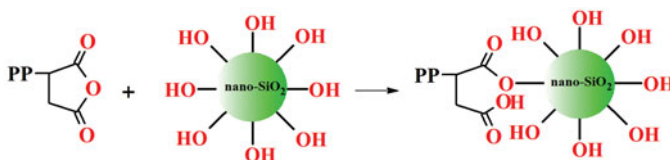


Figure 12. SEM micrographs of (a) PP/EPR/MAH-g-PP/SiO₂, (b) PP/EPR/MAH-g-PP/AEAPTMS/SiO₂ and (c) PP/EPR/MAH-g-PP/MPTMS-SiO₂.

to the uncompatibilized blend. This is attributed to the poor interfacial adhesion between the non-polar PP as well as EPR and the polar MAH-g-PP phases. This shows that the compatibilizer will not be situated at the interface region between the two phases. These results support the conclusions from the mechanical properties findings. Jaziri et al. [54] have also shown that the addition of compatibilizer to (PP/EPR)/EOC system does not induce any reduction in the EPR and EOC sizes. The authors suggest that the MAH-g-PP solubilizes entirely in the PP/EPR phase due to the affinity and the chemical compatibility between PP and MAH-g-PP. The SEM micrographs of the composites are illustrated in Figure 12. It is clear to show that the nano-SiO₂ particles play a dramatic role in reducing the dispersed nodule size in the

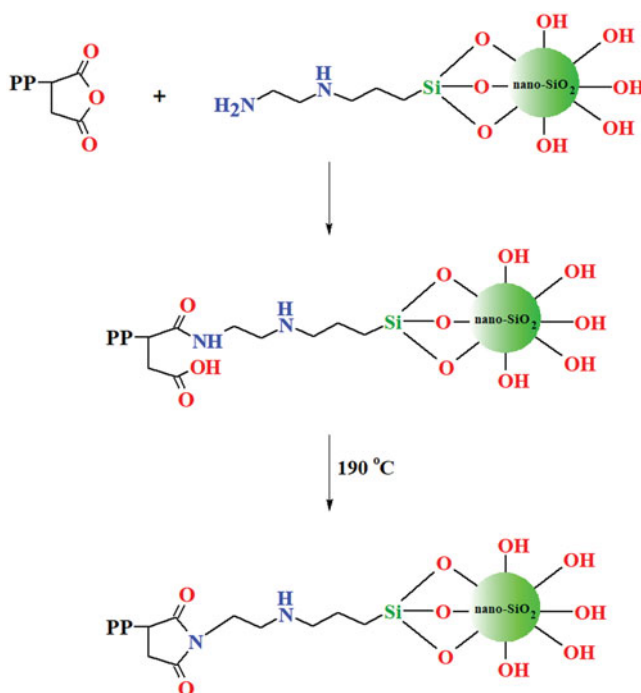


Scheme 1. Chemical reaction between MAH-g-PP and nano-SiO₂.

PP matrix. This decrease in the size of the elastomer suggests the increase of the compatibility between PP and EPR since the decrease of the nodule size in the polymer blend is related to the decrease of the interfacial tension between the two polymers. These results are consistent with the work of Huneault et al. [55], who showed that the addition of MAH-g-PP in a PP/EOC blend causes a reduction of modulus and elongation at break. Doufnoune and Haddaoui [3] investigated the PP/EPR/graphene oxide composites and found that the domain size of EPR phase in the PP matrix also decreased significantly even when MAH-g-PP and graphene oxide were added.

Figure 12(a) presents the SEM micrograph of compatibilized PP/EPR/SiO₂. It is shown that the use of MAH-g-PP has significantly promoted a better dispersion of the nanofiller in the matrix. This indicates that the addition of a polar rubber is favorable for the dispersion in PP composites. This compatibilizer acts as an emulsifier at the interface and thus reduces the interfacial tension leading to a small and uniform distribution of the EPR phase. In addition, the MAH-g-PP can also react with silica nanofiller due to the dipolar interaction between the hydroxyl groups of the silica nanoparticles and maleic anhydride functional groups. As indicated in scheme 1, the combination of MAH-g-PP results in the chemical reaction between the polymer matrix and nano-SiO₂ particles, which in turn improves their interfacial compatibility. Previous studies on three-phase (polymer/elastomer/filler) composites have produced contradictory results in the phase structures in such composites. In some investigations, separate dispersion of the components was observed [18,19], while in others the encapsulation was perceived [20]. However, other studies [21,22] have reported that under certain conditions both structures can be formed.

Figures 12(b)–(c) illustrate the SEM micrographs of the PP/EPR/MAH-g-PP/AEAPTMS-SiO₂. It is shown that the composites prepared using AEAPTMS-SiO₂ show a more pronounced reduction of the domains size of the elastomer phase. On the other hand a better dispersion of the nanofiller was observed due to the reduction of the electrostatic forces between the silica nanoparticles. This corroborates the high tensile strength and Young's modulus that were obtained in the PP/EPR/MAH-g-PP/AEAPTMS-SiO₂ system. Scheme 2 shows the chemical reactions between the amino silane coupling agent (AEAPTMS) and maleic anhydride groups of grafted PP (MAH-g-PP). The primary amino groups, react instantaneously with the carboxyl groups resulting from the opening of the anhydride ring, because of its high susceptibility to hydrolysis [3,33]. The subsequent cyclisation step in which imides are forming is slower and reversible but can be effectively completed during the melt-processing of the composites [48,56,57].



Scheme 2. Chemical reaction between MAH-g-PP and AEAPTMS-SiO₂.

5. Conclusions

Polypropylene/ethylene-propylene rubber (PP/EPR) composites containing silica nanoparticles (nano-SiO₂) were prepared by a melt blending masterbatch process using a Brabender mixer. Interfacial interactions were modified to improve the polymer/filler compatibility and ultimately the mechanical properties of the composites by surface modification of nano-SiO₂ particles with two different silane coupling agents: N-(β -aminoethyl)- γ -aminopropyltrimethoxysilane (AEAPTMS) and 3-methacryloxypropyltri-methoxysilane (MPTMS). In order to achieve a better dispersion of the nanofiller and to enhance its interaction with the polymer matrix, methylmethacrylate grafted PP (MMA-g-PP), glycidymethacrylate grafted PP (GMA-g-PP) and maleic anhydride grafted PP (MAH-g-PP) were used as compatibilizers to mediate the polarity between the nanofiller surface and PP/EPR matrix. The mechanical properties of the blends and composites were determined through tensile and impacts tests. The tensile strength and Young's modulus of the composites were improved in the presence of the silylated nano-SiO₂ and the compatibilizers. A significant improvement of these properties was achieved by the presence of functionalized AEAPTMS-SiO₂ and MAH-g-PP. Conversely, the elongation at break and the impact strength of the composites decreased with respect to neat PP/EPR blend. The SEM study revealed a two-phase morphology in which the elastomer phase appeared as the dispersed phase. The DSC results showed that the introduction of modified silica and MAH-g-PP increased the crystallinity of the composites. The TGA analysis showed that the incorporation of both nano-SiO₂ particles and compatibilizers improved significantly the thermal stability. The SEM results also showed that the addition of MAH-g-PP

into PP/EPR composites contributed to a better dispersion of the EPR phase and nano-SiO₂ particles in the polymer matrix.

Disclosure statement

No potential conflict of interest was reported by the authors.

References

- [1] Mehta S, Mirabella FM, Rufener K, et al. Thermoplastic olefin/clay nanocomposites: morphology and mechanical properties. *J Appl Polym Sci.* 2004;92(2):928–936.
- [2] Lim JW, Hassan A, Rahmat AR, et al. Rubber toughened polypropylene nanocomposite: effect of polyethylene octene copolymer on mechanical properties and phase morphology. *Appl Polym.* 2006;99(6):3441–3450.
- [3] Doufnoune R, Haddaoui N. Effects of surface functionalized partially reduced graphene oxide and different compatibilizers on the properties and structure of PP/EPR nanocomposites. *J Polym Res.* 2017;24:138–144.
- [4] Choudhary NR, Chaki TK, Bhowmick AK. Thermal characterization of thermoplastic elastomeric natural-rubber-polypropylene blends. *Thermo-Chim Acta.* 1991;176:149–161.
- [5] Da Silva ALN, Tavares MIB, Politano DP, et al. Polymer blends based on polyolefin elastomer and polypropylene. *J Appl Polym Sci.* 1997;66:2005–2014.
- [6] Danesi S, Porter RS. Blends of isotactic polypropylene and ethylene propylene rubbers: rheology, morphology and mechanics. *Polymer.* 1978;19:448–457.
- [7] Di Y, Iannace S, Nicolais L. Thermal behavior and morphological and rheological properties of polypropylene and novel elastomeric ethylene copolymer blends. *J Appl Polym Sci.* 2002;86:3430–3439.
- [8] Da Silva ALN, Rocha MCG, Fernanda MB. Study of rheological behavior of elastomer/polypropylene blends. *Polym Test.* 2002;21:289–293.
- [9] Da Silva ALN, Rocha MCG, Fernanda MB, et al. Evaluation of rheological and mechanical behavior of blends based on polypropylene and metallocene elastomers. *Polym Test.* 2002;21:647–652.
- [10] Mae H, Omiya M, Kishimoto K. Comparison of mechanical properties of PP/SEBS blends at intermediate and high strain rates with SiO₂ nanoparticles Vs. CaCO₃ Fillers. *J Appl Polym Sci.* 2008;110:1145–1157.
- [11] Premphet K. Structure-property characterization of ternary phase polypropylene composites [dissertation]. Uxbridge Middlesex: Brunel University of London Kingston Lane; 1995.
- [12] Svab I, Musil V, Pustak A, Smit I. Wollastonite-reinforced polypropylene composites modified with novel metallocene EPR copolymers. II. Mechanical properties and adhesion. *Polym Composites.* 2009;30(7):1091–1097.
- [13] Dubnikova IL, Berezina SM, Antonov AV. The effect of morphology of ternary-phase polypropylene/glass bead/ethylene-propylene rubber composites on the toughness and brittle-ductile transition. *J Appl Polym Sci.* 2002;85:1911–1928.
- [14] D’Orazio L, Mancarella C, Martuscelli E, Sticotti G, Massari P. Melt rheology, phase structure and impact properties of injection-molded samples of isotactic polypropylene/ethylene-propylene copolymer (iPP/EPR) blends: influence of molecular structure of EPR copolymers. *Polymer.* 1993;34:3671–3681.
- [15] Bhadane P. Morphology of ethylene-propylene-diene terpolymer/polypropylene (EPDM/PP) blends and their thermoplastic vulcanizates [dissertation]. École polytechnique de Montréal; 2005.
- [16] Rajkiran R, Tiwari, Paul DR. Polypropylene-elastomer (TPO) nanocomposites: 1. Morphology. *Polymer.* 2011;52(21):4955–4969.

- [17] Mnif N, Massardier V, Kallel T, Elleuch B. New (PP/EPR)/nano- CaCO_3 based formulations in the perspective of polymer recycling. Effect of nanoparticles properties and compatibilizers. *Polym Adv Technol*. 2010;21(12):896–903.
- [18] Jancar I. Dynamic moduli of CaCO_3 and $\text{Mg}(\text{OH})_2$ -filled polypropylene. *J Mat Sci*. 1991;26(15):4123–4129.
- [19] Kolarik I, Lednický F. Structure of polypropylene/EPDM elastomer/calcium carbonate composites. In: Sedlacek B, editor. *Polym composites*. Berlin: Walter de Gruyter; 1986. p. 537–544.
- [20] Comitov PG, Nicolova ZG, Simeonov IS, et al. Compositions à base de polypropylène possédant une résistance au choc améliorée. *Eur Polym J*. 1984;20(4):405–407.
- [21] Pukanszky B. Structure, blends and composites. In: Karger-korcsis J, editor. *Composites, Vol 3, particulate filled polypropylene: structure and properties*. London: Chapman and Hall; 1995. p. 1–70.
- [22] Pukanszky B, Tudos F, Kolarik J, et al. Ternary composites of polypropylene, elastomer, and filler—analysis of phase-structure formation. *Polym Composites*. 1998;11(2):98–104.
- [23] Stamhuis IE. Mechanical properties and morphology of polypropylene composites. III. Short glass fiber reinforced elastomer modified polypropylene. *Polym Composites*. 1988;9(4):280–284.
- [24] Kolarik J, Pukanszky B, Lednický F. Ternary composites polypropylene/elastomer/filler: Phase structure and its control. In: Ishida H, editor. *Interfaces in polymer, ceramics, and metal matrix blends*. New York (NY): Elsevier; 1988. p. 453–462.
- [25] Marosi GY, Bertalan GY, Rusznak I, et al. Effect of additives on the structure and mechanical properties of PP-filler system, In: Sedlacek B, editor. *Polym composites*. Berlin-New York (NY): Walter de Gruyter; 1986. p. 457–464.
- [26] Chiang WY, Yang W, Pukanszky B. Polypropylene composites. II: Structure-property relationships in two- and three-component polypropylene composites. *Polym Eng Sci*. 1992;32(10):641–648.
- [27] Theisen A, Heß M, Kosfeld R. Analysis of ternary composites of PP/EPDM/Filler, In: Lemstra PJ, Kleintjens LA, editors. *Integration of polymer science and technology*. Vol5, Elsevier Appl Sci. London: 1990, p.72–76.
- [28] Mirzadeh A. Thermoplastic vulcanizate nanocomposites based on polypropylene/ethylene-propylene diene terpolymer (PP/EPDM) prepared by reactive extrusion [dissertation]. École polytechnique de Montréal; 2012.
- [29] Li C, Deng H, Wang K, et al. Strengthening and toughening of thermoplastic polyolefin elastomer using polypropylene-grafted multiwalled carbon nanotubes. *J Appl Polym Sci*. 2011;121(4):2104–2112.
- [30] Li C, Zhao Q, Deng H, et al. Preparation, structure and properties of thermoplastic olefin nanocomposites containing functionalized carbon nanotubes. *Polym Int*. 2011; 60(11):1629–1637.
- [31] Hausmann K, Flaris V. Polymeric coupling agents as property enhancers in highly filled polymer systems. *Polym Polym Comp*. 1997;5(2):113–119.
- [32] Doufnoune R, Haddaoui N, Riahi F. The effects of coupling agents on the mechanical, rheological and thermal properties of calcium carbonate-filled LDPE compatibilized with maleic anhydride-g-LDPE (part II). *Int J Polym Mater*. 2008;57(7):690–708.
- [33] Doufnoune R, Haddaoui N, Riahi F. Elaboration and characterization of an organic/inorganic hybrid material: effect of the interface on the mechanical and thermal behavior of PP/ CaCO_3 composites. *Int J Polym Mater*. 2006;55(10):815–834.
- [34] Doufnoune R, Haddaoui N, Riahi F. Effect of coupling agents on the performance of PP/MAH-g-PP/ CaCO_3 ternary composites. *Int J Polym Mater*. 2006;55:989–1007.
- [35] Uotila R, Hippí U, Paavola S, et al. Compatibilization of PP/elastomer/microsilica composites with functionalized polyolefins: Effect on microstructure and mechanical properties. *Polymer*. 2005;46(19):7923–7930.
- [36] Chen JH, Rong MZ, Ruan WH, et al. Interfacial enhancement of nano- SiO_2 /polypropylene composites. *Compos Sci Technol*. 2009; 69(2):252–259.

- [37] Pustak A, Denac M, Leskovac M, et al. Polypropylene/silica micro-and nanocomposites modified with poly(styrene-*b*-ethylene-co-butylene-*b*-styrene). *J Appl Polym Sci.* 2015; 132(6):41486–41498.
- [38] Pustak A, Denac M, Leskovac M, et al. Morphology and mechanical properties of iPP/silica composites modified with (styrene-*b*-ethylene-co-butylene-*b*-styrene) grafted with maleic anhydride. *Polym-Plast Technol Eng.* 2015;54:647–660.
- [39] El-Midany AA, Ibrahim SS. The interfacial role of compatibilizers to improve mechanical properties of silica–polypropylene composites. *Physicochem Probl Miner Process.* 2011;46(1):295–305.
- [40] Brandrup J, Immergut EH, Grulke EA, et al, editors. *Polymer handbook*. 4th ed. New York (NY): John Wiley and Sons; 2005.
- [41] Monredon-Senani S. Interaction organosilanes/silice de precipitation du milieu hydro-alcoolique au milieu aqueux [dissertation]. Paris (FR): Université de Paris VI; 2004.
- [42] Migliorini S. Agent de couplage et surfaces modèles de silice, suivi en infrarouge ATR du greffage d'organosilanes sur oxyde de silicium [dissertation]. Montpellier (FR): Université de Montpellier II; 2000.
- [43] Chiang C H, Ishida H, Koenig JL. The structure of γ -aminopropyltriethoxysilane on glass surfaces. *J Colloid Interf Sci.* 1980;74(2):396–404.
- [44] Doufnoune R, Haddaoui N, Riahi F. The Interactions of silane and zirconate coupling agents with calcium carbonate. *Inter J Polym Mater.* 2007;56(3):227–246.
- [45] Bellamy LJ. *The infrared spectra of complex molecules*. New York (NY): John Wiley and Sons; 1954.
- [46] Tam WY, Cheung T, Li RKY. An investigation on the impact fracture characteristics of EPR toughened polypropylene. *Polym Test.* 1996;15(4):363–379.
- [47] van der Wal A, Nijhof R, Gaymans RJ. Polypropylene rubber blends 2: the effect of the rubber content on the deformation and impact behaviour. *Polymer.* 1999;40(22): 6031–6044.
- [48] Doufnoune R. Contribution à l'étude du comportement des mélanges Polymères/charges en présence d'agents de couplage [dissertation]. Sétif (ALG): Université Ferhat Abbas de Sétif 1; 2005.
- [49] Sichina WJ. Prediction of end-use characteristics of polyethylene materials using differential scanning calorimetry USA: Application Briff DSC-11; 1994.
- [50] Tjong, SC, Xu SA, Yu Li RK, et al. Preparation and performance characteristics of short-glass-fiber/maleated styrene-ethylene-butylene-styrene/polypropylene hybrid composites. *J Appl Polymer Sci.* 2002;86(5):1303–1311.
- [51] Xu W, Liang G, Zhai H, et al. Preparation and crystallization behaviour of PP/PP-g-MAH/Org-MMT nanocomposite. *Eur Polym J.* 2003;39(7):1467–1474.
- [52] Guezout Z, Doufnoune R, Haddaoui N. Effect of graphene oxide on the properties of compatibilized polypropylene/ethylene-propylene-rubber blend. *J Polym Res.* 2017;24: 129–143.
- [53] Dong Q, Ding Y, Wen B, et al. Improvement of thermal stability of polypropylene using DOPO-immobilized silica nanoparticles. *Colloid Polym Sci.* 2012;290:1371–1380.
- [54] Jaziri M, Mnif N, Massardier-Nageotte V, et al. Rheological, thermal, and morphological properties of blends based on poly(propylene), ethylene propylene rubber, and ethylene-1-octene copolymer that could result from end of life vehicles: effect of maleic anhydride grafted poly(propylene). *Polym Eng Sci.* 2007;47(7):1009–1015.
- [55] Huneault MA, Godfroy PG, Lafleur PG. Performance of talc/ethylene-octene copolymer/polypropylene blends. *Polym Eng Sci.* 1999;39(6):1130–1138.
- [56] Demjén Z, Pukanszky B, Nagy J. Possible coupling reactions of functional silanes and polypropylene. *Polymer.* 1999;40(7):763–1773.
- [57] Sánchez-Valdes S, Méndez-Nonell J, Medellín-Rodríguez FJ, et al. Evaluation of different amine-functionalized polyethylenes as compatibilizers for polyethylene film nanocomposites. *Polym Int.* 2010;59(5):704–711.